



ELECTRICAL AND COMPUTER ENGINEERING

To: Dr. Newton

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Section Number: CPE 360

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Subject: Lab 0: Equipment Familiarization

Introduction

The goal of this lab is to familiarize ourselves with the lab equipment we will be using this semester.

Task 1

Three Resistors, a 1k, 10k, and 100k, were selected and placed in series, in that order. 10V was supplied via the power supply.

	Resistance	Voltage (Calculated)	Voltage (Measured)	Current (Calculated)	Current (Measured)
V1	1k Ω	10 V	10.0062 V	0.9009 mA	0.912 mA
V2	10k Ω	9.9099 V	9.9157 V	0.9009 mA	0.912 mA
V3	100k Ω	9.0090 V	9.0126 V	0.9009 mA	0.912 mA

Current measurements were consistent since it was the same loop, while voltage dropped slowly due to the 100k Ω resistor at the end of the circuit.

A text based version of the circuit can be seen below

VCC –V1–1k Ω –V2–10k Ω –V3–100k Ω –GND

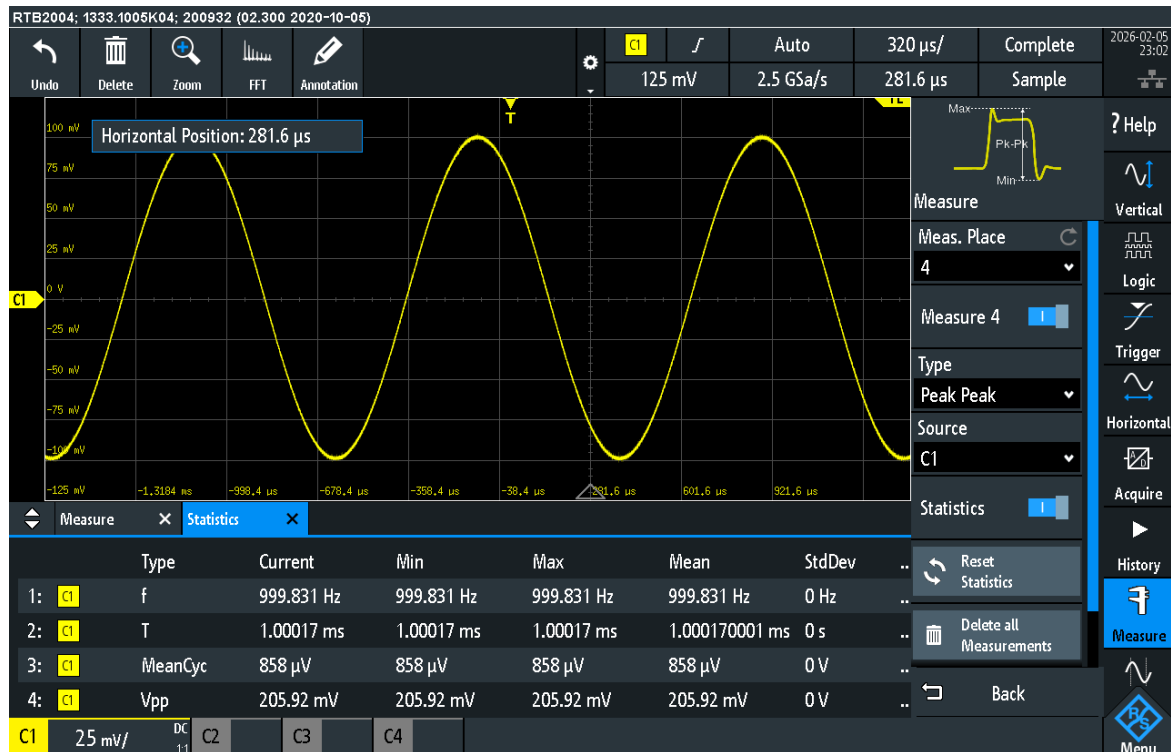
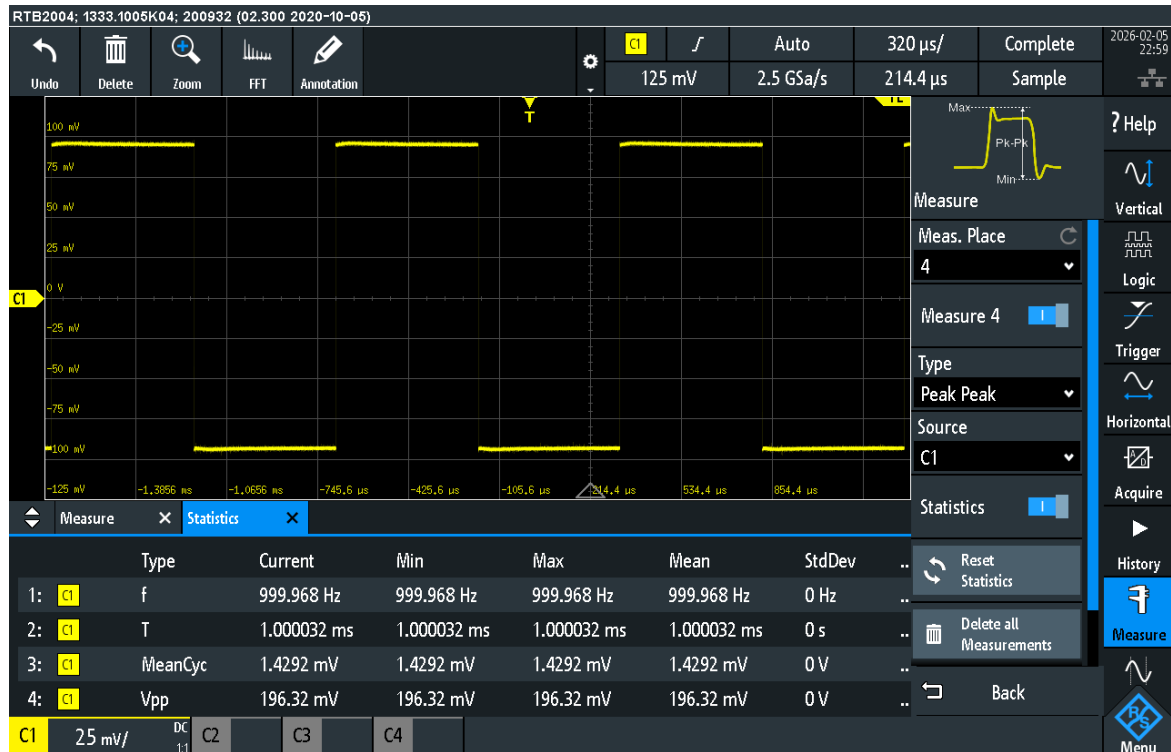
Task 2

Three Resistors, a 1k, 10k, and 100k, were selected and placed in parallel. 10V was supplied via the power supply.

	Resistance	Voltage (Calculated)	Voltage (Measured)	Current (Calculated)	Current (Measured)
V1	1k Ω	10 V	10.0057 V	10 mA	10.1582 mA
V2	10k Ω	10V	10.0057 V	1 mA	1.0202 mA
V3	100k Ω	10V	10.0057 V	0.1 mA	0.1013 mA

Current measurements were consistent since it was the same loop, while voltage dropped slowly due to the 100k Ω resistor at the end of the circuit.

Task 3



Task 4:

The following code was written to blink the LED. It uses the delay.h header to toggle the built in LED ever 500 ms, for a cycle of 1 second. While only Pin 5 of Port B is used to control the LED, it was easier to just toggle the whole port, while making it so I can use the same code for Task 5 with little to no effort spent on finding the right pin.

```
#define F_CPU 16000000
#include <avr/io.h>
#include <util/delay.h>

int main(int argc, char** argv) {

    DDRB = 0xFF;
    while (1) {
        PORTB ^= 0xFF;
        _delay_ms(500);
    }

}
```

The function operates on the assumption that the board is running at it's max speed of 16 MHz. Once you define the clock speed correctly, it operates with a period of exactly one second.

Task 5

Because the Task 4 code was made to affect the entire Port B, a connector was placed into a pin on Port B, another was placed to ground, then a 220 ohm resistor and red LED was placed in series to connect them. After aligning the LED properly, it blinked at the same speed as the rest of the on board LED.